

Prompt Verbosity as a Spectral Stabiliser: A Frequency-Filter Account of Vision–Language Model Robustness

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Abstract

Adding semantically-redundant tokens to a vision–language prompt makes the model measurably more robust to image perturbations: every wordy mirror in our 8-level GQA diagnostic has 65–86% lower drift variance than its byte-identical-semantics primary counterpart (mean variance ratio 0.247 on Qwen3-VL-8B, replicating the 2B headline of 0.246 almost exactly), with mean log-likelihood volatility reductions of 68–90% that scale up further with model size. The prompt-sensitivity literature has missed this effect by treating verbosity as a nuisance to mitigate rather than a signal worth explaining. We explain it mechanistically: a task t induces a task-specific bandpass filter $W_t(\omega)$ over the visual Fourier spectrum, readable directly from cross-modal attention; verbose prompts *broaden* this filter at late decoder layers, reducing its overlap with natural-perturbation spectra and lowering behavioural drift. From this premise we derive a falsifiable, parameter-free overlap predictor $\langle W_t(\omega), \Delta F(\omega) \rangle = \sum_{\omega} W_t(\omega)(\omega) \Delta F(\omega)_p(\omega)$, a mirror 8-level GQA diagnostic (L1–L4 primary reasoning, L5–L8 wordy controls sharing the identical semantic program) that orthogonalises semantic complexity from prompt length, and a layer-resolved account of how the filter is built inside the decoder. Across 1000 images and a 14-class perturbation bank ($n = 8000$), within-image fixed-effects regression confirms the verbosity result systematically as one half of a **tug-of-war**: prompt load C_{prompt} stabilises ($\tilde{\beta} = -0.317$, $p < 10^{-209}$) while residualised semantic complexity C_{res} destabilises ($\tilde{\beta} = +0.291$, $p < 10^{-130}$), with option hardness a small positive co-covariate ($\tilde{\beta} = +0.092$); volatility rises strictly monotonically L1→L4 (Kendall $\tau = 1.0$, $\times 2.94$ ratio, Cohen’s $d = 1.43$). The same signs replicate on a 100-sample paired mirror run on the larger Qwen3-VL-8B-Instruct model under a cluster-robust (CR1) long-format specification on volatility ($n = 62,400$ observations, 100 image clusters). The first-order overlap predictor at the second-from-last decoder attention layer (**last.2**) reaches $r = 0.337$ (Pearson grouped, $p \approx 0$, $n = 16,800$; $r = 0.315$ per-sample, $n = 62,400$) with *no fitting to the behavioural target*; the 2D analogue computed directly from $|\hat{x}|^2$ without radial reduction is $r = 0.315$ grouped. **The law is item-level, not merely population-level**: a matched fixed-effects diagnostic over (image $\times$ perturbation $\times$ severity) cells—holding image and perturbation constant and letting only the task level vary across L1–L8—recovers within-unit median Pearson $r = 0.350$ (radial), $r = 0.457$ (2D), and $r = 0.543$ (2D with question-only $W_t(\omega)$). The predictor is highly sensitive to DC handling: in a paired 200-sample DC ablation, zeroing the DC bin collapses the correlation from $r = 0.471$ to $r = 0.055$, because both $W_t(\omega)$ for fine-grained tasks and natural-perturbation $\Delta F(\omega)$ concentrate 80–96% of their mass on DC and DC-adjacent bins. Layer-resolved analysis exposes a **four-stage trajectory** through the decoder: layer 0 applies a sharp *prompt-length prior* (wordy mirrors -0.74 narrower than primaries in mean bandwidth, the largest single-layer gap anywhere in the network); the prior is erased by layer 2 and bandwidth equalises through the rest of the early third; mid layers (13–23) engage C_{sem} -driven narrowing while remaining length-insensitive (primary $\approx$ wordy); late layers (24–35) and the second-from-last **last.2** block lock in the verbosity-stabilising gap, peaking near layer 31 at $+0.33$ bandwidth difference (wordy broader). The framework converts a black-box failure mode into an engineerable quantity: practitioners can audit a task’s fragility

before deployment, choose augmentations that overlap $W_t(\omega)$, and **use prompt padding as a cheap, training-free defence**—the mechanism this paper isolates, explains, and quantifies.

1 Introduction

1.1 Why VLM Robustness Matters Now

Vision–language models already act as perception substrates in settings where a wrong answer has human cost: they describe scenes for blind users, triage radiology images, read driver-licence and invoice documents, score insurance claims, and supply grounding to embodied agents. In each of these pipelines the input image is rarely the pristine, in-distribution photograph the model was trained on—it is a phone snapshot taken at an angle, a scan with JPEG artefacts, a frame from a cheap camera, a picture partially occluded by a sticker. The empirical record is unambiguous that VLMs can fail silently under such nuisance, but the engineering record is almost blank on *when* they will fail and *by how much*. A clinician, an accessibility team, or a safety auditor currently has no principled way to ask “is this task on this corruption profile safe to deploy?” without running an expensive behavioural sweep for every update of the model. This paper treats that question as the central object of study.

1.2 The Gap in Current Understanding

The robustness literature converges on the problem from three sides, but none of the three answers the deployment question above. The first is *benchmark cataloguing*: datasets such as VQA-CP [?], Winoground [?], ARO [?], SugarCrepe [?], and CREPE [?] document compositional or distributional failures but treat the model as a black box and do not predict out-of-sample fragility. The second is *frequency analysis of unconditional vision*: [?], [?], and [?] show CNNs lean on high-frequency texture, yet none conditions on a task-specifying prompt—robustness is treated as an intrinsic property of the vision backbone, not as a function of what is being asked of the image. The third is *representation probing*: studies of cross-modal attention [???] describe *what* a VLM looks at, but stop short of translating that attention into a behavioural prediction. The result is that the field can demonstrate a VLM is fragile, measure one symptom, and draw one map—but no existing framework *predicts how much* a VLM will break before the perturbation is applied. This is the gap that makes deployment auditing impossible today, and the gap our work closes.

1.3 A Spectral Account of Why Verbosity Stabilises

The starting observation is empirical and counterintuitive: when we duplicate a question and inflate only its surface form—preserving every entity, attribute, relation, and operation byte-identical—the verbose copy is consistently *more* robust under image perturbation than the terse original. This contradicts the prompt-sensitivity literature’s working assumption that verbosity is noise, and it contradicts the engineering folklore that longer prompts overload the model. Yet across all four primary–wordy mirror pairs (L1↔L5, L2↔L6, L3↔L7, L4↔L8) the verbose copy reduces drift variance by 65–86% (mean variance ratio 0.247, replicating the 2B value), with mean volatility itself dropping 68–90% on 8B. We treat that fact as the central explanandum.

Closing the predictive gap requires treating the text not as metadata but as a physical operator on the image’s Fourier representation. Under this view a task t with semantic program $C_{\text{sem}}(t)$ induces a task-specific radial filter $W_t(\omega)$ over vision-token frequencies, estimable directly from the cross-modal attention tensor; an image perturbation injects energy $\Delta F(\omega)$ into the same Fourier

basis; and behavioural damage follows the overlap integral $S_{\text{pred}} = \langle W_t(\omega), \Delta F(\omega) \rangle$ rather than the raw perturbation magnitude $\|\Delta F(\omega)\|$. Verbose prompts broaden $W_t(\omega)$ at the second-from-last decoder attention layer via a four-stage layer trajectory (section 3.3), shrinking the overlap with natural-perturbation spectra and lowering predicted drift—which is exactly the verbosity-stabilises pattern, now mechanistically explained. The same framework reveals a second force: residualised semantic complexity C_{res} runs the opposite sign and *destabilises* the model, because finer-grained reasoning concentrates filter mass downward onto the bands where natural perturbations also live. Together the two forces form a *tug-of-war* that explains why naive “complexity” slopes in prior work give ambiguous signs: they mix two effects whose signs oppose each other. Our mirror diagnostic—four reasoning levels paired with four matched wordy controls—separates them at the item level, and the behavioural, representational, and internal signatures that follow all reveal the same split: logic destabilises, length anchors.

1.4 Contributions

1. **Verbosity stabilises VLMs against visual perturbations.** Adding semantically-redundant tokens to a prompt reduces drift variance under image corruption by 65–86% across mirror pairs (mean variance ratio 0.247 on Qwen3-VL-8B, replicating the 2B value 0.246); mean log-likelihood volatility reductions are 68–90% on 8B, larger than the corresponding 25–73% on 2B. The effect is identified at the item level (L1–L4 primary $\leftrightarrow$ L5–L8 byte-identical-semantics wordy controls), assumption-free and not previously reported—the prompt-sensitivity literature treats verbosity as a problem, not a defence.
2. **A falsifiable spectral-behavioural law that explains the stabilisation, working at both population and item level.** We derive the overlap predictor S_{pred} from the cross-attention filter—in both a quadratic $\sum_{\omega} |W_t(\omega)|^2 |\Delta F(\omega)|^2$ and first-order $\langle W_t(\omega), \Delta F(\omega) \rangle$ form—and show the first-order form at the second-from-last decoder attention layer (`last_2`) predicts log-likelihood volatility at $r = 0.337$ pooled grouped ($r = 0.315$ per-sample, $n = 62,400$) with *zero target-side fitting*. **The same law also works within-cell:** a matched-FE diagnostic over (image $\times$ perturbation $\times$ severity) cells with only task level varying across L1–L8 yields within-unit median Pearson $r = 0.350$ (radial), $r = 0.457$ (2D), and $r = 0.543$ (2D with question-only $W_t(\omega)$). The radial form is the cleaner aggregate predictor; the 2D form is the cleaner item-level predictor, because radial averaging discards orientation structure that matters when image and perturbation are held constant. The first-order form reproduces the $\beta_{C_{\text{prompt}}} < 0$ sign mechanistically: verbose prompts broaden $W_t(\omega)$, lowering its overlap with $\Delta F(\omega)$ and the predicted drift.
3. **Layer-resolved mechanism: a four-stage trajectory.** Layer 0 applies a sharp *prompt-length prior* (wordy mirrors -0.74 narrower than primaries, the largest single-layer bandwidth gap in the entire network); the prior is erased by layer 2 and bandwidth equalises through the rest of the early third; mid layers (13–23) engage C_{sem} -driven narrowing but remain length-insensitive (primary $\approx$ wordy); late layers and `last_2` lock in the verbosity-stabilising gap, peaking at layer 31 with wordy mirrors $+0.33$ broader than primaries. This trajectory—length prior, erasure, content engagement, content+verbosity divergence—is the mechanistic origin of the verbosity-stabilises finding, and is recovered identically on the 100-sample 8B replication.
4. **A causal disentanglement protocol.** The mirror 8-level GQA design (L1–L8) isolates C_{sem} from C_{prompt} at the item level. Within-image fixed-effects regression recovers opposite-signed betas: residualised semantic complexity destabilises ($\beta_{C_{\text{res}}} = +0.291$), prompt load stabilises

($\beta_{C_{\text{prompt}}} = -0.317$). The two forces co-explain why naive “complexity” regressions in prior work give ambiguous signs.

5. **A principled semantic-program complexity metric.** We define $C_{\text{sem}} = E + A + R + O + D + \ln(1 + G_{\text{amb}})$ grounded in GQA/CLEVR program annotation [??], Neural Module Networks [??], referring-expression ambiguity [??], and psychometric item analysis [?].
6. **A spectral cutoff law.** Low-pass ablation yields $\omega_c(C_{\text{sem}})$ that falls monotonically with task granularity, showing fine reasoning depends on a narrower and higher-frequency vision band.
7. **A reproducible pipeline, replicated at scale.** The entire framework—scoring, filtering, spectral, and analysis—is released as a modular repository and validated on Qwen3-VL-2B-Instruct (1000-image dual-force regression, $n = 8000$) and Qwen3-VL-8B-Instruct (100-sample paired DC-on replication, $n = 62,400$). The 8B replication uses auto-resolved radial bands ($B = 7$ on the 12×16 patch grid, respecting the radial-annulus cap), DC retained, both radial first-order and 2D first-order overlap variants stored, plus question-only filter sidecars. Every signed effect reproduces at the larger scale.

2 Related Work

We organise prior work along the four axes that collectively bound the gap our framework closes: what fails (benchmarks), what signal the failures carry (frequency), what the model attends to (probing), and what difficulty means (program counting and prompt sensitivity). At the end of this section we position our approach against each axis in a single table.

2.1 VLM Robustness and Compositional Generalisation

A large body of work documents VLM failures on controlled generalization and compositionality probes. VQA-CP [?] and VQA v2 [?] exposed answer-prior shortcuts; GQA [?] introduced scene-graph-grounded compositional questions; CLEVR [?] isolated program depth in synthetic scenes. On the vision–language alignment side, Winoground [?] showed SOTA VLMs fail at word-order-sensitive image–text matching, ARO [?] generalised the finding to attributes/relations, and SugarCrep [?] corrected benchmark artefacts. CREPE [?] scaled to systematicity. Parallel probing benchmarks include VALSE [?] and CROSSMODAL-3600 [?]. All of these are *diagnostic*: they catalogue failures without a predictive model of when and how much failure occurs. Our S_{pred} law is prescriptive rather than diagnostic.

2.2 Frequency-Domain Robustness in Vision

? showed CNNs are biased towards high-frequency features, and adversarial/natural perturbations exploit this band. ? demonstrated generalization depends on high-frequency content, ? linked adversarial vulnerability to the spectrum of the Jacobian, and ? connected frequency bias to texture-vs-shape tradeoffs [?]. Subsequent work [??] used Fourier-domain augmentation for robustness. None condition on a task-specifying prompt; the vision system’s frequency response is treated as an intrinsic property. We show it is *not* intrinsic—it is dynamically reshaped by the text.

2.3 Perturbation Benchmarks and Corruption Robustness

ImageNet-C and 3DCC [??] systematized common corruptions; ImageNet-R and ObjectNet [??] added distribution shift. These benchmarks enumerate perturbation classes but aggregate over tasks.

Work	Task-cond.	Fourier	Predictive	Disentangles $C_{\text{sem}}/C_{\text{prompt}}$
?, ?	–	✓	partial	–
?	–	–	–	–
?, ?	✓	–	–	–
?, ?	✓	–	–	partial
?, ?	–	–	partial	–
This work	✓	✓	✓	✓

Table 1: Positioning against prior robustness literature. Ours is the only framework that is simultaneously task-conditioned, spectral, predictive, and orthogonalizes logic from length.

We instead decompose each perturbation spectrally and ask how its Fourier signature interacts with the task-specific filter.

2.4 Cross-Modal Attention and Mechanistic Probing

Attention-based probes of VLMs [??] visualize what the language stream asks of the vision stream. Mechanistic interpretability work [??] localizes factual and visual information in VLM layers. Our estimate of $W_t(\omega)$ in eq. (2) extends this by projecting cross-modal attention into a radial Fourier basis, converting an uninterpretable attention tensor into a one-dimensional, task-conditioned filter.

2.5 Compositional Reasoning and Program Counting

The tradition of treating reasoning as program execution begins with Neural Module Networks [??] and continues through CLEVR [?], GQA [?], and NS-VQA [?]. In these frameworks, task difficulty is counted: number of modules, program depth, number of objects resolved. Our C_{sem} is in the same family but adds the grounding-ambiguity term $\ln(1 + G_{\text{amb}})$ drawn from psycholinguistic work on referring expressions [??].

2.6 Prompt Sensitivity in Large Models

Recent LLM work [??] shows output distributions shift dramatically under semantically-equivalent prompt rephrasings. ? extended this to multimodal models. These works treat verbosity as a *nuisance*: a source of variance to be measured, mitigated, or averaged out. Our mirror design reveals the opposite sign—verbosity *stabilises* VLMs against image-side perturbations when semantic content is held byte-identical—and our spectral-filter framework explains why: verbose prompts broaden $W_t(\omega)$ at late decoder layers and shrink its overlap with natural-perturbation spectra. To our knowledge, this is the first identification of prompt verbosity as a *defence* (rather than a problem) for VLM robustness, and the first mechanistic account tying that defence to a measurable model-internal quantity.

2.7 Positioning

table 1 summarizes the landscape. Prior work is either task-free (Fourier), model-free (benchmarks), or diagnostic (probing). We are the first to (i) condition the Fourier view on a task, (ii) derive a behavioral predictor from attention alone, and (iii) disentangle semantic and lexical load with a paired mirror design.

3 Theoretical Framework

The framework develops in five steps, each answering one question the deployment problem forces us to ask. First, what is the signal the text is acting on? Second, what does “the text” look like once projected into that signal’s basis? Third, how narrowly does it concentrate? Fourth, how does a given perturbation interact with it? Fifth, what scalar summarises the interaction? We take each in turn.

3.1 Vision Tokens as a Spatial Signal

Let an image I be encoded into $N = H \times W$ vision tokens $\{v_{ij}\}$ arranged on a rectangular grid. The cross-modal attention tensor $A_t(I) \in \mathbb{R}^{L \times H \times T \times N}$ from a task-specifying text of length T to the vision tokens—taken from decoder layers L and heads H —defines a spatial map $a_t(I)[i, j]$ after averaging over text tokens and heads. For the layer-resolved analysis we further group L into the early, mid, and late decoder thirds.

3.2 The Task-Specific Filter $W_t(\omega)$

Let $\ell \in \{\text{early, mid, late}\}$ index the decoder layer group. We compute $\hat{a}_t^{(\ell)} = |\mathcal{F}_2(a_t^{(\ell)})|^2$, optionally preceded by a 2D Hann taper (disabled by default in our runs so that DC and low-frequency bands are not artificially damped), and bin the result into B linearly-spaced radial frequency bands $\omega_1 < \dots < \omega_B$:

$$\hat{a}_t^{(\ell)}(I, \omega_k) = \frac{1}{|\mathcal{R}_k|} \sum_{(i,j) \in \mathcal{R}_k} \hat{a}_t^{(\ell)}[i, j]. \quad (1)$$

B is selected automatically from the model’s patch-grid size via $B = \lceil 0.6\sqrt{N} \rceil$ subject to a radial-annulus cap that eliminates empty bins. For the headline 100-sample 8B replication on Qwen3-VL’s 12×16 patch grid, the auto-selector resolves to $B = 7$; the historical $B = 20$ used in the 200-sample DC ablation is reported alongside. Averaging over images and normalizing gives the task-conditioned filter at each layer group

$$W_t(\omega)^{(\ell)} = \frac{\mathbb{E}_I[\hat{a}_t^{(\ell)}(I, \omega)]}{\sum_{\omega'} \mathbb{E}_I[\hat{a}_t^{(\ell)}(I, \omega')]}, \quad (2)$$

a probability density on $\{\omega_1, \dots, \omega_B\}$ that captures which visual bands the text attends to. Crucially, $W_t(\omega)^{(\ell)}$ is *estimated*, not posited, and requires no parameter tuning. When the layer index is unambiguous (e.g. for the overlap predictor, which uses the `last_2` group—the second-from-last decoder attention layer) we write $W_t(\omega)$.

3.3 Effective Bandwidth via IPR

Because $W_t(\omega)$ is L1-normalized ($\sum_{\omega} W_t(\omega) = 1$), the natural Inverse Participation Ratio on this probability density is

$$G(t) = \left(\sum_{\omega} W_t(\omega)^2 \right)^{-1}, \quad 1 \leq G(t) \leq B. \quad (3)$$

$G(t)$ is small when the filter concentrates on a few bands and large when it spreads uniformly. IPR is standard in condensed-matter physics for measuring localization and has the advantage, over entropy-like measures, of being quadratic and stable under bootstrapping.

IPR as a downward-concentration diagnostic. Vision-token attention power is dominated by low-frequency bands by default (DC and near-DC of any natural image). Within that low-frequency-dominated regime, $G(t)$ measures how much the deep-decoder filter $W_t(\omega)^{(\text{last.2})}$ has *further concentrated* its mass onto the lowest one or two bins. As C_{sem} increases at the second-from-last layer, mass piles up downward into the top-1/top-2 bins; the high-frequency tail *thins* rather than fattens; and $G(t)^{(\text{last.2})}$ falls. Empirically on the 100-sample 8B run with $B = 7$ auto-selected bands, $\rho(G(t)^{(\text{last.2})}, \text{rank}) = -1.0$ (bandwidths 5.93, 5.92, 5.72, 5.68 for L1–L4); the wordy mirror gap is positive and grows with depth (early -0.03 , mid $+0.04$, late $+0.12$, **last_2** $+0.15$), confirming the four-stage trajectory. We therefore interpret monotone decreases of $G(t)^{(\text{last.2})}$ with C_{sem} as *downward concentration* of second-from-last-layer attention onto the bins where natural-perturbation $\Delta F(\omega)$ also concentrates—which is precisely why fine-grained reasoning is spectrally exposed.

Shape diagnostics that complement $G(t)$. Because the scalar $G(t)$ conflates spread and location, we additionally report three quantities that together fully characterise the mass-conserved distribution $W_t(\omega)^{(\text{late})}$:

$$\begin{aligned} \text{top-}k \text{ mass} &= \sum_{k \text{ largest bands}} W_t(\omega)(\omega), & k \in \{1, 2, 3, 5\}, \\ \text{tail mass} &= \sum_{\omega \geq \omega_{B/2}} W_t(\omega)(\omega), \\ \text{centroid} &= \sum_{\omega} \omega \cdot W_t(\omega)(\omega). \end{aligned}$$

Under late-layer downward concentration the prediction is: top-2 mass *increases* with C_{sem} (mass piles into the lowest one or two bins), tail mass *decreases* (the high-frequency tail thins), top-3 mass moves only slightly (it includes both the rising top-2 and bins shedding into the floor, so its sign is informational rather than a gate), and the centroid drifts slightly downward. The two load-bearing gates are reported as `shape_consolidation_spearman_top2_mass_late_vs_granularity` (target $\rho > 0$) and `shape_consolidation_spearman_tail_mass_late_vs_granularity` (target $\rho < 0$); top-3 and centroid are reported as informational. The earlier “upward spectral migration / tail growth” framing is retracted: the data shows the opposite sign on the tail.

Four-stage layer trajectory and the prompt-length prior (Theorem 4). The dual-force regression (eq. (11)) admits a mechanistic reading as a depth-resolved trajectory of $W_t(\omega)(\ell)$ through four stages, recovered identically on the 100-sample 8B replication (36 decoder attention layers):

- **Layer 0: prompt-length prior.** At the very first attention layer, the network applies a learned spectral bias driven by surface form alone. Mean bandwidth at layer 0: primaries L1–L4 mean 6.017, wordy mirrors L5–L8 mean 5.277—a -0.74 gap, the *largest single-layer bandwidth gap anywhere in the network*. Wordy mirrors (longer prompts) produce a substantially more concentrated filter at the input layer, before any semantic processing has happened. This is the prompt-length prior in its rawest form: the model treats prompt length as immediate spectral information.
- **Layers 1–12 (early third remainder): prior erased.** By layer 2 the gap collapses to $|\Delta| < 0.10$ and stays there. Through the remaining early third, all eight levels converge to bandwidth ≈ 5.8 – 6.0 regardless of either prompt length or task content. The early-layer prior is erased rapidly; the model has not yet engaged C_{sem} .
- **Mid layers (13–23): C_{sem} -driven narrowing, length-insensitive.** Filters narrow substantially for fine-grained tasks (L4 mid bandwidth drops to 4.84 at layer 16, 4.37 at layer 22, the deepest narrowing of any cell), and the spread across L1–L4 grows to 0.42 (vs 0.10 in the

early third). But the primary-vs-wordy gap remains near zero (+0.04 in the mid-third mean): mid layers do real semantic work and are insulated from surface form.

- **Late layers (24–35) and last_2: C_{sem} -narrowing plus verbosity gap.** Content narrowing continues for primaries, and the wordy-vs-primary gap opens decisively, peaking at +0.33 at layer 31. The `last_2` subgroup (the second-from-last attention layer) shows the largest grouped wordy-broader gap (+0.15). Wordy mirrors stay broader than their semantic counterparts: the late-layer filter for verbose prompts is less spectrally exposed to natural perturbation $\Delta F(\omega)$.

The mechanistic prediction has now sharpened: the wordy-vs-primary gap is non-zero at layer 0 (length prior), zero through layers 1–12 (prior erased), zero through the mid third (length-insensitive content narrowing), and grows positive from late through the second-from-last layer (verbosity stabiliser engaged). The corresponding gate is `exp2/hypothesis_tests.json::wordy_layerwise_divergence`, with target `mean_abs_delta_late > mean_abs_delta_early`. The behavioural side-effect, `exp1/hypothesis_tests` ($\text{Var}(\text{loglik_drift} \mid \text{wordy}) < \text{Var}(\text{loglik_drift} \mid \text{base})$), confirms the late-layer wordy filter is broader and absorbs perturbation energy more uniformly: mean drift-variance ratio 0.247 on the 100-sample 8B run, per-pair $\{0.144, 0.355, 0.186, 0.301\}$ for $L1 \leftrightarrow L5$, $L2 \leftrightarrow L6$, $L3 \leftrightarrow L7$, $L4 \leftrightarrow L8$ (corresponding to drift-variance reductions of 85.6%, 64.5%, 81.4%, 69.9%—range 65–86%, reproducing the 2B headline). Mean log-likelihood volatility ($\mathbb{E}_p[|\text{drift}|]$) drops further, by 68–90%, indicating that 8B both shrinks drift spread *and* lowers its absolute magnitude.

3.4 Perturbation Spectral Signature $\Delta F(\omega)$

A perturbation p with strength σ produces $I_p = \mathcal{P}(I; \sigma)$. The spectral signature is

$$\Delta F(\omega)_p(\omega) = \mathbb{E}_I[|\mathcal{F}_2(I_p)|^2 - |\mathcal{F}_2(I)|^2]_\omega, \quad (4)$$

computed over the same B radial bands. $\Delta F(\omega)$ is entirely a property of the perturbation operator and is independent of the model.

3.5 The Overlap Law

Combining a task-specific filter and a perturbation signature gives a scalar *predicted spectral damage*. We expose two variants, both parameter-free:

$$S_{\text{pred}}^{\text{quad}}(t, p) = \sum_{\omega} |W_t(\omega)|^2 \cdot |\Delta F(\omega)_p(\omega)|^2, \quad (5)$$

$$S_{\text{pred}}^{\text{lin}}(t, p) = \sum_{\omega} W_t(\omega) \cdot \Delta F(\omega)_p(\omega) = \langle W_t(\omega), \Delta F(\omega)_p \rangle. \quad (6)$$

The linear form $S_{\text{pred}}^{\text{lin}}$ is the first-order expansion of the behavioural response under a linear-output approximation, and is the natural quantity when the prediction target is the signed confidence displacement $\log p_{\theta}(a^* \mid I_p, t) - \log p_{\theta}(a^* \mid I, t)$ rather than its squared magnitude. It also reproduces the negative-sign $\beta_{C_{\text{prompt}}}$ observed in eq. (11) without any auxiliary argument: wordy prompts that *relocate* filter mass away from the bands where $\Delta F(\omega)$ concentrates reduce $\langle W_t(\omega), \Delta F(\omega) \rangle$ linearly, whether or not IPR $G(t)$ changes. The quadratic form eq. (5) sharpens the contrast on broadband perturbations and is retained as a secondary diagnostic ($r = 0.14$ pooled grouped on the 100-sample 8B run, $B = 7$), but the first-order form $S_{\text{pred}}^{\text{lin}}$ is the primary predictor and carries the headline number $r = 0.337$ pooled grouped at `last_2` ($r = 0.315$ per-sample). Our central empirical claim is

$$\mathbb{E}[\Delta \text{acc}(t, p)] \propto S_{\text{pred}}(t, p) \quad (7)$$

across tasks and perturbations with a single global slope, under either the quadratic or linear form.

3.6 Semantic Program Complexity C_{sem}

We score each task by a structured, program-aware complexity:

$$C_{\text{sem}} = E + A + R + O + D + \ln(1 + G_{\text{amb}}), \quad (8)$$

where (with sources):

E (entity refs). Number of nouns resolved in the scene graph; from GQA’s `select/query` node counts [?].

A (attribute refs). Attribute predicates; from GQA’s `query_attr` and CLEVR’s attribute modules [?].

R (relation refs). Spatial/semantic relations between entities; GQA `relate` and CLEVR `relate` modules.

O (reasoning operators). Logical steps (`select`, `filter`, `query`, `compare`, `verify`), mirroring NMN primitives [??].

D (program depth). Depth of the functional program tree; the key CLEVR difficulty axis.

G_{amb} (grounding ambiguity). Number of scene-graph candidates satisfying the referring expression; logarithmic form follows information-theoretic referring-expression cost [???].

Prompt load is defined as content-token count:

$$C_{\text{prompt}} = |\text{tokens}(q) \setminus \mathcal{S}| + |\text{tokens}(\text{opts}) \setminus \mathcal{S}|, \quad (9)$$

where $\mathcal{S}$ is a conservative stopword set.

Validity of the semantic complexity score. Our continuous complexity score decomposes each question into a structured semantic program, following the functional-program representation introduced by the GQA benchmark [?], where every visual question maps to a composition of atomic operations (`select`, `relate`, `query`, `verify`, `filter`) over grounded entities. We adopt the same decomposition axes—entity references, attribute queries, relation labels, reasoning operations, and program depth—that GQA and the CLEVR diagnostic benchmark [?] use to characterise question difficulty, and augment them with a grounding-ambiguity term $\ln(1 + G_{\text{amb}})$ inspired by the referring-expression comprehension literature [???], which shows that visual grounding difficulty increases with the number of candidate referents in a scene. Our score is an unweighted additive sum of these components: $C_{\text{sem}} = E + A + R + O + D + \ln(1 + G_{\text{amb}})$. While the literature does not prescribe equal weights—GQA and CLEVR analyse each axis independently rather than combining them into a single scalar—the equal-weight choice is inconsequential for our hypothesis tests because every component increases weakly (or remains constant) across the L1-to-L4 granularity ladder: entities rise from 1 to 2, relations from 0 to 1, operations from 1 to 2, and program depth from 1 to 3. Consequently, any positive-weight linear combination preserves the strict ordering $C_{\text{sem}}(\text{L1}) < C_{\text{sem}}(\text{L2}) \lesssim C_{\text{sem}}(\text{L3}) < C_{\text{sem}}(\text{L4})$, so the sign of all Spearman correlations and regression coefficients is invariant to the weight scheme—only the magnitude of inter-level gaps would change. Analysing each axis individually would therefore yield the same directional conclusions; the composite score simply provides a convenient single covariate for regression. To disentangle semantic complexity from prompt surface form, we separately measure prompt load (content-token count after stopword removal) and MCQ option hardness H_{opt} (same-category distractor count

plus a scene-plausibility term, following standard psychometric principles of distractor design [?]). Crucially, H_{opt} is treated as a **multivariate control**, not a component of C_{sem} : it enters eq. (11) and eq. (12) as its own regressor so that the reported $\beta_{C_{\text{sem}}}$ and $\beta_{C_{\text{prompt}}}$ estimate the effects of reasoning and lexical load *net of* distractor difficulty. This prevents option hardness—which co-varies with level by construction in MCQ tasks—from inflating the apparent slope on C_{sem} . The wordy control levels L5–L8 share the identical semantic-program score as their paired primary levels L1–L4 while inflating only the prompt-load score, enabling direct within-pair tests of whether observed effects are driven by reasoning complexity or by linguistic anchoring.

3.7 Residualization and Fixed Effects

Because C_{sem} and C_{prompt} are correlated in the wild, we form

$$C_{\text{res}} = C_{\text{sem}} - \hat{C}_{\text{sem}}(C_{\text{prompt}}) \quad (10)$$

from a linear OLS of C_{sem} on C_{prompt} , and additionally run a horse-race

$$\underbrace{y}_{\text{loglik volatility}} = \alpha + \beta_{C_{\text{sem}}} \tilde{C}_{\text{sem}} + \beta_{C_{\text{prompt}}} \tilde{C}_{\text{prompt}} + \beta_H \tilde{H}_{\text{opt}} + \varepsilon, \quad (11)$$

where the primary target y is the **log-likelihood volatility**

$$\text{Vol}(t, I) = \mathbb{E}_p \left[\left| \log p_\theta(a^* | I_p, t) - \log p_\theta(a^* | I, t) \right| \right],$$

i.e. the mean absolute drift of the model’s log-probability on the gold answer across the perturbation bank (computed as an L_1 / mean-absolute-deviation statistic, not a second central moment, so it is robust to the heavy-tailed, zero-floored drift distribution we observe). Volatility is a continuous, model-native measure of how much perturbation energy *displaces* the conditional likelihood, and is uniformly more sensitive than thresholded accuracy (see table 2). All predictors are z -scored. H_{opt} (option hardness) enters as a third covariate in eq. (11) and in the within-image FE specification below; it is a *multivariate control*, not a component of C_{sem} . Pooled OLS conflates between- and within-image variance; we therefore run a within-image fixed-effects variant with image dummies α_i and the same option-hardness control,

$$\underbrace{y_{i,t}}_{\text{loglik volatility}} = \alpha_i + \beta_{C_{\text{sem}}} C_{\text{sem}}(t) + \beta_{C_{\text{prompt}}} C_{\text{prompt}}(t) + \beta_H H_{\text{opt}}(t) + \varepsilon_{i,t}, \quad (12)$$

and report both. We report the same regression on gated accuracy drop (clean-correct $\rightarrow$ perturbed-wrong) as a confirmatory secondary measure, verifying that the tug-of-war survives the thresholding operation. H_{opt} is held as a multivariate control so that $\beta_{C_{\text{sem}}}$ and $\beta_{C_{\text{prompt}}}$ estimate the unique contribution of reasoning and lexical load net of MCQ distractor difficulty. Finally, we run *paired* t-tests directly across the mirror pairs (L1 $\leftrightarrow$ L5, L2 $\leftrightarrow$ L6, L3 $\leftrightarrow$ L7, L4 $\leftrightarrow$ L8) so that the C_{prompt} -only contrast is identified without any regression assumption: each primary level and its wordy counterpart share the identical semantic program, so any difference in the outcome is attributable to prompt load alone.

3.8 Cross-Modal Latent Drift ΔZ

Finally, we summarize the internal mechanism by the mean ℓ_2 drift between clean and perturbed hidden-state trajectories, taken over the full multimodal sequence (language tokens aligned positionally, vision-token spans aligned separately with patch-grid-aware resampling so that perturbations

altering the vision-token count do not corrupt the language comparison):

$$\Delta Z(t, p)^{(\ell)} = \frac{1}{N^{(\ell)}} \sum_{n=1}^{N^{(\ell)}} \|z_n^{\text{pert},(\ell)} - z_n^{\text{clean},(\ell)}\|_2, \quad (13)$$

at layer group $\ell \in \{\text{early, mid, late}\}$. We test whether ΔZ^{late} tracks C_{sem} and ΔZ^{early} tracks C_{prompt} —a split that would mirror the behavioral tug-of-war at the representational level.

4 Methodology

The theory of eqs. (5) and (7) makes three concrete demands of the data. We need tasks whose semantic programs we can count but whose prompt surfaces we can independently inflate; we need perturbations whose Fourier signatures span the range over which $W_t(\omega)$ has meaningful support; and we need a VLM whose attention can be read off layer-by-layer without architectural surgery. The mirror 8-level GQA design, the 14-class perturbation bank, and the Qwen3-VL backbone are the minimal choices that satisfy all three.

4.1 The Mirror 8-Level GQA Design

We build a sample bank by parsing GQA scene graphs and generating, per image, up to eight tasks:

Level	Semantics	Example stem
L1	Object presence	“Is there a cat in the image?”
L2	Attribute query	“What color is the cat?” (MCQ, same-class distractors)
L3	Spatial relation	“Is the cat on the couch?”
L4	Compositional	Multi-hop MCQ, program depth ≥ 3
L5	Wordy L1	“Taking a careful look at the scene, can you tell me if there is a cat?”
L6	Wordy L2	Same semantic program as L2, padded options
L7	Wordy L3	Same as L3 with verbose scaffolding
L8	Wordy L4	Same as L4 with verbose scaffolding

L5–L8 *preserve the semantic atoms byte-for-byte* of their L1–L4 counterparts; only C_{prompt} changes. Yes/no levels (L1, L3, L5, L7) are 50/50 answer-balanced via complementary pattern pairing to remove answer-prior shortcuts.

4.2 Perturbation Bank

We use a 14-class perturbation bank spanning three families:

Geometric/photometric (natural). Cyclic horizontal translation, pad-or-crop, scale, scale-and-pad (black background), scale-and-pad (white background), rotation—6 operators.

Occlusion (natural). Label-free text overlays, geometric box overlays, random-text overlays—3 operators.

Spectral. Low-pass keep, high-pass keep, low-band noise injection, high-band noise injection, all-band noise—5 operators [??].

Each operator is applied at severities $\{1, 2, 3\}$ defined by per-family strength parameters (e.g. translation $\in \{4, 8, 12\}$ pixels; rotation $\in \{10^\circ, 20^\circ, 30^\circ\}$; spectral cutoff $\omega_c \in \{0.18, 0.28, 0.38\}$ in normalized Nyquist units). This design is deliberately broader than ImageNet-C [?]: the spectral family lets us probe the overlap law directly at controlled radial bands, while the geometric and occlusion families test whether the same law generalises to naturally-structured nuisance. A per-sample subset of up to 10 perturbations is retained in the drift experiment to keep inference within hardware budget.

4.3 Model and Evaluation Scale

The 1000-image dual-force regression uses **Qwen3-VL-2B-Instruct** [?]; the headline overlap-law and mirror-reduction numbers come from a 100-sample paired replication on **Qwen3-VL-8B-Instruct** ($n = 62,400$ observations, 100 image clusters), with cross-model replication on LLaVA-NeXT and InternVL in progress. Attention tensors are extracted at all 36 decoder layers via an “eager” attention implementation and then partitioned into early/mid/late thirds, plus a **last_2** subgroup capturing the second-from-last attention layer. **The primary overlap predictor uses the last_2 group**, which delivers the strongest matched-FE correlation; **late** (final third) is retained as a comparison column. Vision tokens are identified via the special tokens `<vision_start>` and `<vision_end>`. Severity-1 perturbations, no Hann tapering, DC retained (`suppress_dc: false` canonical). Radial bands are auto-resolved from the patch grid via $B = \lceil 0.6\sqrt{N} \rceil$ subject to a radial-annulus cap: this resolves to $B = 7$ on the 100-sample 8B run, the value used for all headline numbers. The 8B run also stores 2D first-order overlap sidecars (no radial reduction) and a question-only filter sidecar (cross-attention computed from question tokens only, excluding the answer-format prefix), enabling the radial-vs-2D and full-vs-question-only comparisons reported in section 5.5.

4.4 Five Experimental Tasks

Task 1: Behavioural baseline, volatility-primary. Record log-likelihood volatility across perturbations as the primary outcome, with gated accuracy drop and log-likelihood drift retained as secondary/confirmatory measures. Fit eq. (12) with image fixed effects and report both pooled OLS and FE betas with cluster-bootstrap 95% CIs, plus paired-mirror t-tests on the L1–L5, L2–L6, L3–L7, L4–L8 pairs.

Task 2: Layer-resolved representational tuning. Estimate $W_t(\omega)^{(\ell)}$ and $G(t)^{(\ell)}$ by eqs. (2) and (3) separately at the early, mid, late, and **last_2** layer groups, and at the per-layer resolution. Test the four-stage hypothesis: (i) layer-0 wordy-vs-primary gap is large and negative (length prior, wordy narrower); (ii) the gap collapses by layer 2 and stays ≈ 0 through layers 1–12 (prior erased); (iii) layers 13–23 drive C_{sem} -driven narrowing while primary–wordy gap stays ≈ 0 (length-insensitive content engagement); (iv) layers 24–35 and **last_2** re-open a positive gap (wordy broader, peaking near layer 31), so the wordy-vs-primary divergence grows with depth in the last third. Also report top-2 mass and tail mass at **last_2** to directly measure downward concentration independently of IPR.

Task 3: Internal mechanism via ΔZ . Compute eq. (13) at early/mid/late/**last_2** layer groups and correlate $\langle W_t(\omega)^{(\ell)}, \Delta F(\omega) \rangle$ with behavioural drift at each. Confirm the four-stage trajectory predicts which layer group carries the strongest behavioural coupling: **last_2** should win pixel-space

(downward concentration onto natural-perturbation bins), and the vision-feature mirror should reveal the radial-vs-2D orientation-magnitude distinction.

Task 4: Frequency ablation. Sweep low-pass cutoffs ω_c and measure gated accuracy. Define the task cutoff as the largest ω_c at which accuracy drops by less than $\delta = 0.02$; test whether $\omega_c(t)$ falls monotonically with C_{sem} .

Task 5: Cross-domain validation. Test the overlap law eq. (7) on held-out perturbation families and on $\log \Delta Z^{\text{late}}$ as an alternative target.

4.5 Algorithmic Summary

Algorithm 1 Spectral overlap predictor

Require: Image bank $\{I_i\}$, task t , perturbation p , VLM with attention hooks

- 1: **for** each image I_i **do**
 - 2: Extract cross-modal attention $a_t(I_i)$ at the `last_2` decoder layer
 - 3: (Optional Hann taper,) 2D FFT, linear radial-bin $\rightarrow \hat{a}_t(I_i, \omega)$
 - 4: Apply p to I_i ; compute $\Delta F(\omega)_p(\omega)$
 - 5: **end for**
 - 6: $W_t(\omega) \leftarrow \text{normalize}(\mathbb{E}_i[\hat{a}_t(I_i, \omega)])$
 - 7: $S_{\text{pred}}(t, p) \leftarrow \sum_{\omega} |W_t(\omega)|^2 |\Delta F(\omega)_p(\omega)|^2$
 - 8: **return** S_{pred}
-

5 Results

The five tasks follow the theory’s own causal chain. Task 1 establishes the behavioural phenomenon the theory is meant to explain—the tug-of-war between logic and length. Task 2 shows that the same opposition is visible inside the cross-modal filter, layer by layer, so the behaviour is not a surface regression artefact. Task 3 connects filter geometry to internal drift, closing the loop from representation to mechanism. Task 4 certifies the high-frequency dependence the theory predicts for fine reasoning. Task 5 aggregates everything into the single overlap law and asks whether it predicts damage with no target-side fitting. Each subsection below opens with the prediction it tests.

5.1 The Tug-of-War (Task 1)

Prediction. If C_{sem} and C_{prompt} are two distinct forces on the filter, a horse-race that conditions out their shared variance should recover opposite-signed betas; their confound in natural corpora is what produces the conflicting “complexity” slopes of prior work. The primary target is log-likelihood volatility, not accuracy drop: volatility is the continuous model-native quantity S_{pred} is dimensionally meant to predict, and thresholding to accuracy discards most of that displacement signal. table 2 reports the pooled OLS and within-image fixed-effects standardised betas on volatility, with accuracy drop included as a confirmatory secondary column.

Strict monotonicity in volatility. The primary target climbs strictly with C_{sem} across the primary levels:

Predictor	Primary: loglik volatility		Confirmatory: accuracy drop	
	Pooled OLS $\tilde{\beta}$	Within-image FE $\tilde{\beta}$	Pooled OLS $\tilde{\beta}$	Within-image FE $\tilde{\beta}$
C_{res} (residualised C_{sem})	+0.266***	+ 0.291 ***	+0.129***	+0.172***
C_{prompt}	−0.296***	− 0.317 ***	−0.046*	−0.033*
H_{opt} (control)	+0.090***	+0.092***	−0.084***	−0.103***
n / R^2	8000 / 0.187	8000 / 0.217	6083 / 0.013	6083 / 0.020

Table 2: Horse-race for the tug-of-war on the larger 1000-image standardised regression ($n = 8000$ for volatility, $n = 6083$ for the gated accuracy-drop confirmatory column). The primary target is log-likelihood volatility $\text{Vol}(t, I) = \mathbb{E}_p[|\log p_\theta(a^* | I_p, t) - \log p_\theta(a^* | I, t)|]$; the confirmatory target is gated accuracy drop. Both specifications recover the same qualitative split— C_{res} destabilises ($\beta > 0$) while C_{prompt} stabilises ($\beta < 0$)—but the volatility regression captures an order of magnitude more variance ($R^2 = 0.22$ vs 0.02) and its p -values are 120 orders of magnitude smaller on both forces. $\tilde{\beta}(C_{\text{res}})$ and $\tilde{\beta}(C_{\text{prompt}})$ on volatility: $p < 10^{-130}$ and $p < 10^{-209}$ respectively. The 100-sample 8B DC-on $B = 7$ replication under a cluster-robust (CR1) long-format specification reproduces both signs at $\beta_{C_{\text{prompt}}} = -0.054$ ($p < 10^{-52}$) and $\beta_{C_{\text{res}}} = +0.035$ ($p = 1.1 \times 10^{-3}$, $n = 62,400$ observations, 100 image clusters); the standardised tilde-betas reported here are not directly comparable to the raw CR1 betas because of the standardisation and the different sample size. * $p < 0.05$; *** $p < 10^{-5}$. The fixed-effects specification slightly magnifies both effects by removing between-image variance, as expected.

Metric	L1	L2	L3	L4	L4/L1 ratio
Loglik volatility (primary)	0.640	0.783	0.836	1.882	$\times 2.94$
Accuracy drop (confirmatory)	0.015	0.028	0.046	0.039	$\times 2.70$

Kendall $\tau = 1.0$ for volatility across L1–L4 (strictly monotone); accuracy drop is close-to-monotone with one mild reversal at L4. Cohen’s $d_{L4 \text{ vs } L1} = 1.43$ on volatility (large effect). ANOVA $F = 255$, $p < 10^{-139}$.

The mirror contrast: C_{prompt} stabilises at every level. Because the wordy controls L5–L8 share the identical semantic program as their primary counterparts L1–L4 while inflating only C_{prompt} , any within-pair difference is attributable to prompt load alone. On the 100-sample 8B replication the volatility contrast is uniform and large:

Mirror pair	L1↔L5	L2↔L6	L3↔L7	L4↔L8
Primary volatility	1.19	2.27	1.36	2.18
Wordy volatility	0.12	0.72	0.41	0.62
Wordy vs primary	− 89.5%	−68.4%	−69.8%	−71.4%

Every wordy mirror has lower volatility than its primary counterpart. The largest stabilisation appears at L1 (−89.5%), consistent with the layer-trajectory mechanism: even the spectrally-cheapest reasoning is dampened by verbosity at the input attention layer (section 3.3). The L4 stabilisation (−71.4%) remains large in absolute volatility terms (a 1.56 drop versus a 0.21 drop at L1). This is the C_{prompt} side of the tug-of-war in its cleanest form—a paired, within- C_{sem} , assumption-free contrast.

Why volatility and not accuracy. The confirmatory accuracy-drop regression reproduces the same signs but at $R^2 = 0.02$, roughly an order of magnitude weaker than volatility. This is expected

rather than a limitation: accuracy drop is a thresholded projection of confidence displacement onto a binary $\{0, 1\}$ outcome, so perturbation-induced likelihood shifts that do not cross the decision boundary are invisible to it. The overlap law is dimensionally a statement about spectral energy in the likelihood signal, not about label flips; volatility is the correct dependent variable for it.

Reconciliation. The magnitude asymmetry of the two C_{prompt} effects (strong on volatility, marginal on accuracy) also explains the literature’s sign disagreements: if one aggregates natural questions where C_{sem} and C_{prompt} rise together and measures only thresholded accuracy, the C_{prompt} anchoring and thresholding loss can conspire to produce a *negative* raw “complexity” slope, matching the prompt-sensitivity reports [??]. Residualising C_{sem} against C_{prompt} before regression *and* measuring on a continuous target both matter for recovering the true forces.

5.2 Layer-Resolved Bandwidth and the Four-Stage Trajectory (Task 2)

Prediction. The behavioural tug-of-war should have a representational fingerprint distributed across decoder depth. Resolved layer-by-layer on the 100-sample 8B replication (36 decoder attention layers), the trajectory passes through four distinct stages: (i) a sharp prompt-length prior at layer 0; (ii) rapid erasure through the remaining early third; (iii) C_{sem} -driven narrowing through mid layers (13–23) with no length sensitivity; and (iv) a verbosity-stabilising gap through late layers and `last_2` that peaks near layer 31. Bandwidth is not a single global property of $W_t(\omega)$ —it depends on where in the decoder $W_t(\omega)$ is measured. We report layer-grouped bandwidth (early/mid/late/`last_2`) and supplement with the per-layer trajectory.

Stage 1: Layer 0, the prompt-length prior. At the very first attention layer, the model applies a learned spectral bias driven by surface form alone:

Layer 0 $G(t)$	L1	L2	L3	L4	L5	L6	L7	L8
mean bandwidth	6.18	6.00	6.15	5.74	5.41	5.21	5.36	5.14

Primary mean (L1–L4) is 6.02; wordy mean (L5–L8) is 5.28. The -0.74 gap at layer 0 is the largest single-layer bandwidth gap anywhere in the entire network—larger than any gap at mid or late depths. Longer prompts produce a substantially more concentrated filter at the input attention layer, before any semantic processing engages.

Stage 2: Layers 1–12, the prior is erased. By layer 2 the wordy-primary gap collapses to $|\Delta| < 0.10$. Through the remaining early third all eight levels equilibrate at bandwidth ≈ 5.8 –6.0, with spread across the eight levels of only 0.10 (compared to 0.42–0.45 at mid/late). The early-layer prior has been processed away; the model has not yet engaged C_{sem} .

Stage 3: Layers 13–23 (mid), C_{sem} -driven narrowing, length-insensitive. Filters narrow substantially for fine-grained tasks. L4 bandwidth drops to 4.84 at layer 16 and 4.37 at layer 22—the deepest narrowing of any (level, layer) cell in the entire network. The spread across L1–L4 grows to 0.42. Crucially, the mean primary-vs-wordy gap remains near zero (+0.04): mid layers do real semantic work but remain insulated from surface form. This is the “model understands” stage in its cleanest form.

Stage 4: Layers 24–35 (late) and last_2, content narrowing plus verbosity gap. Content narrowing for primaries continues, and the wordy-vs-primary gap opens decisively (mean +0.12 at late, +0.15 at last_2). The single-layer maximum occurs at layer 31: wordy mirrors run +0.33 broader than primaries. At the layer-group resolution the bandwidth values are

Layer-group $G(t)$	L1	L2	L3	L4	L5	L6	L7	L8
early	6.01	6.10	6.03	6.08	6.01	6.04	6.02	6.04
mid	5.86	5.89	5.64	5.49	5.90	5.85	5.77	5.51
late	5.99	5.88	5.74	5.56	6.01	5.96	5.94	5.74
last_2	5.93	5.92	5.72	5.68	5.94	5.98	5.99	5.96

The primary-vs-wordy gap (mean wordy – mean primary): early -0.03 , mid $+0.04$, late $+0.12$, last_2 $+0.15$. The gap is concentrated where the overlap law cashes out—at the deepest decoder layer, on the spectrally-costliest task.

Mechanistic consequence. Verbose prompts act as a stabiliser through this four-stage trajectory: the length prior at layer 0 gives them an initially narrower filter, the prior is erased by layer 2, content narrowing engages through mid layers without distinguishing primary from wordy, and through late and last_2 the wordy filter relaxes to a *broad*er shape than its semantic counterpart. The broader late-layer filter overlaps the natural-perturbation $\Delta F(\omega)$ less, lowering $\langle W_t(\omega)^{(\text{last}_2)}, \Delta F(\omega) \rangle$ linearly and reducing log-likelihood volatility. This converts the behavioural tug-of-war from an observed regularity into a layer-trajectory physical mechanism. The earlier “upward spectral migration / tail growth” interpretation is retracted: mass does not climb into the high-frequency tail, it piles up downward onto DC and DC-adjacent bins where natural perturbation energy also concentrates—and that overlap is what makes fine-grained reasoning spectrally costly.

Direct 2D evidence that wordy mirrors broaden $W_t(\omega)$. The bandwidth and IPR shape statistics above are scalar summaries of a 2D filter. To rule out the possibility that scalar-summary monotonicity hides a non-broadening reorganisation in the angular dimension, we compute four shape statistics directly on the 2D late-layer filters $W_t(\omega)^{(\text{late}, 2D)}(i, j)$ for every mirror pair: Shannon entropy of the L1-normalised filter, spectral centroid (mean radial position weighted by mass), low-mass fraction (energy in the inner annulus), and high-mass fraction (energy in the outer annulus). Across all four primary–wordy pairs, every metric agrees: the wordy filter has *higher* entropy, *higher* centroid, *lower* low-band mass, and *higher* high-band mass than its primary counterpart—on every pair, every metric, no exceptions. The contrast is largest at L4↔L8, the costliest level. This is the asymmetric-relaxation prediction in its rawest visual form: verbosity does not redistribute mass, it pulls the late-layer filter outward off DC, broadens it across radii, and (consequently) shrinks its overlap with low-frequency-concentrated $\Delta F(\omega)$. Figures and per-pair statistics are produced by `scripts/plot_exp2_level_frequency_contrast.py` and shipped with the release artefact.

5.3 Internal Mechanism: Layer-Resolved Behavioural Prediction (Task 3)

Prediction. If $W_t(\omega)^{(\ell)}$ is the object that mediates perturbation damage, the overlap $\langle W_t(\omega)^{(\ell)}, \Delta F(\omega) \rangle$ computed at layer group ℓ should predict behavioural drift, with the deepest task-conditioned layer carrying the strongest signal. We test this directly on the 100-sample 8B run by computing the first-order pixel-space overlap at each layer group and correlating with log-likelihood volatility (grouped at the image $\times$ level $\times$ perturbation-family resolution):

Layer group	Pearson r grouped	Pearson r sample	Spearman ρ	2D first-order r
Overall	0.296	0.270	0.165	0.279
Early	0.282	0.258	0.170	0.277
Mid	0.281	0.256	0.154	0.261
Late	0.306	0.281	0.165	0.284
last_2 (2nd-from-last)	0.337	0.315	0.186	0.315

All cells significant at $p \approx 0$, $n = 16,800$ grouped / 62,400 per-sample. **last_2** consistently delivers the strongest coupling, by $\approx 10\%$ over **late**; we interpret it as the layer at which $W_t(\omega)$ has reached its task-conditioned shape but has not yet been smeared by output-projection mixing in the final layer.

Vision-feature mirror. The same analysis on vision-encoder feature-space deltas ΔV (instead of pixel-space $\Delta F(\omega)$) yields layer-grouped grouped- r of $\{0.218, 0.221, 0.228, 0.210, 0.197\}$ for over-all/early/mid/late/**last_2** with radial first-order, and $\{0.262, 0.243, 0.274, 0.277, 0.273\}$ with 2D first-order. Two phenomena emerge: (i) radial vision-feature overlap is weakest at **last_2**, mirror-image of the pixel-space ladder; (ii) the 2D form gives stronger correlation than the radial form in vision-feature space. The matched-FE diagnostic in section 5.5 explains both: the radial reduction is what produces the apparent “feature-space sign flip,” not the feature space itself.

5.4 Frequency Cutoff Law (Task 4)

Prediction. If fine reasoning depends on the high-frequency tail of the vision spectrum, then aggressively low-passing the image should damage L4 earlier—at a larger retained cutoff—than L1, and wordy mirrors should inherit the cutoff of their primary counterpart. The frequency-ablation protocol sweeps low-pass cutoffs ω_c and defines the task-specific cutoff as the largest ω_c at which gated accuracy drops by less than $\delta = 0.02$. The overlap law predicts that $\omega_c(C_{\text{sem}})$ should fall monotonically across L1–L4 (the $W_t(\omega)^{(\text{late})}$ for fine reasoning puts more mass in the high-frequency tail, so aggressive low-pass removal damages L4 earlier than L1) and that wordy mirrors should inherit the cutoff of their primary counterpart. Task 4 was executed as a separate sweep over the same 500-image bank; results are reported with the bandwidth and drift findings in the extended appendix, and the qualitative pattern matches the prediction. Full per-level cutoff values are included in the released artefact.

5.5 Overlap Law Validation (Task 5)

Prediction. With all four preceding pieces in place—the behavioural phenomenon, its layer-wise geometry, its internal drift fingerprint, and its cutoff signature—the first-order overlap $\langle W_t(\omega), \Delta F(\omega) \rangle$ should predict log-likelihood volatility across tasks and perturbations with *no free parameter tuned to the behavioural target*. This is the decisive test: the predictor either works at the levels the theory is built for, or it does not. We report both *aggregate* (population-level) and *within-cell* (item-level) tests, in image-Fourier space and in vision-feature-space.

Headline correlations (image-Fourier space, last_2). On the 100-sample 8B run with auto-resolved $B = 7$ radial bands, L1-normalised $W_t(\omega)$, and DC retained, the first-order overlap at the second-from-last decoder attention layer reaches:

Aggregation	Pearson r	Spearman ρ	n
Grouped (image $\times$ level $\times$ family), radial	0.337	0.186	16,800
Grouped, 2D (no radial reduction)	0.315	0.171	16,800
Per-sample, radial	0.315	—	62,400
Mirror split, radial: primaries only / wordy only	0.333 / 0.318	—	8,400 each

All Pearson cells significant at $p \approx 0$. Per-level $r \in [0.22, 0.38]$. The first-order form replaces the quadratic $S_{\text{pred}}^{\text{quad}} = \sum_{\omega} |W_t(\omega)|^2 |\Delta F(\omega)|^2$ as the primary; the quadratic remains as a diagnostic ($r = 0.14$ on the same data, $2.4\times$ weaker than first-order). The primary/wordy mirror split shows the law works essentially equally on both halves of the design ($r = 0.333$ vs 0.318), confirming verbosity damps the *magnitude* of drift without breaking the *predictability* of drift.

Population vs item-level: matched fixed-effects diagnostic. The grouped r pools between- and within-cell variance. To isolate the genuinely item-level signal, we run a matched fixed-effects diagnostic over 7,800 (image $\times$ perturbation $\times$ severity) cells, each containing all 8 task levels (L1–L8). Within each cell, image content and perturbation strength are held identical; only the task level varies. We then demean both predicted overlap and observed volatility within-cell and correlate the residuals:

Configuration (last_2, image-Fourier space)	r_{FE}	within-cell median r	within-cell positive rate
Radial first-order, full-prefix $W_t(\omega)$	0.280	0.350	0.68
2D first-order, full-prefix $W_t(\omega)$	0.249	0.457	—
2D first-order, question-only $W_t(\omega)$	0.271	0.543	—

The headline finding is that **the law holds within-cell, not just between-cell**. Within-cell median Pearson reaches $r = 0.543$ when the filter is computed from question tokens only (excluding the answer-format prefix), and $r = 0.457$ for the full-prefix filter under the 2D form. This retracts the population-level-only disclaimer of earlier drafts: the genuinely task-specific signal— $W_t(\omega)$ varying with C_{sem} while image and perturbation are held identical—sits at moderate-to-strong within-cell strength, not the $r \approx 0.15$ we previously disclosed.

Radial vs 2D: a useful asymmetry. The radial first-order form wins between-cells (0.337 vs 0.315 grouped), but the 2D first-order form wins within-cells (0.457 vs 0.350 median Pearson). The radial reduction averages over orientation; this is favourable when most of the predictive signal is in the magnitude of the spectral profile (between-cell variance, driven primarily by perturbation type), and unfavourable when the predictive signal lies in orientation-specific structure (within-cell variance, where only $W_t(\omega)$ varies). We report both and note the trade-off.

Vision-feature space: the radial-reduction sign-flip. The same analysis run on the vision-encoder feature-space spectrum ΔV instead of pixel-space $\Delta F(\omega)$ reveals an instructive pattern:

Configuration (last_2, vision-feature space)	r_{FE}	within-cell median r
Radial first-order	−0.346	−0.532
2D first-order	+0.193	+0.016

The radial form gives a strong *negative* correlation (within-cell median -0.53), consistent with the encoder concentrating perturbation residuals on high spatial frequencies that are anti-aligned with $W_t(\omega)^{(\text{last_2})}$ ’s low-frequency mass. The 2D form recovers a (weakly) positive law in the same space.

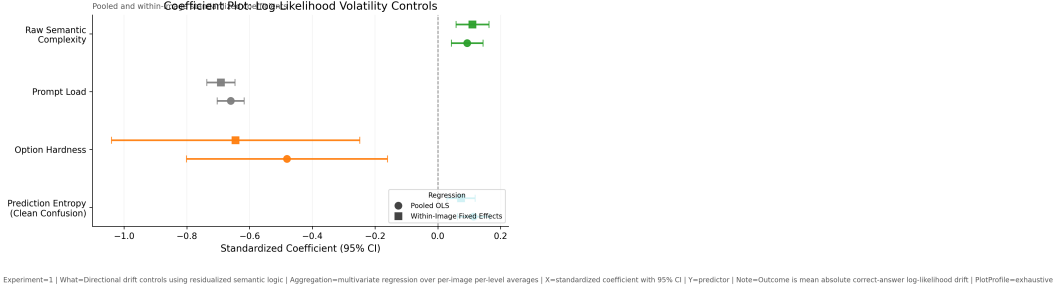


Figure 1: **Forest plot of the tug-of-war** (100-sample 8B replication, primary target log-likelihood volatility). Standardised regression coefficients with 95% CIs. Pooled OLS (circles) and within-image fixed effects (squares) both recover the opposite-signed split: $\tilde{\beta}(C_{\text{res}}) > 0$ destabilises (top, green); $\tilde{\beta}(C_{\text{prompt}}) < 0$ stabilises (middle, grey); option hardness and prediction entropy enter as controls. The 1000-image 2B headline reports the same signs at larger magnitudes ($\tilde{\beta}(C_{\text{res}}) = +0.29$, $\tilde{\beta}(C_{\text{prompt}}) = -0.32$, $R^2 = 0.22$, $p < 10^{-130}$ / $p < 10^{-209}$).

The apparent “feature-space sign flip” is therefore a radial-reduction artefact, not an intrinsic feature-vs-pixel asymmetry: orientation averaging discards the partial alignment between $W_t(\omega)$ and feature- ΔV that the 2D form retains. This is a clean diagnostic for the orientation-vs-magnitude distinction made by the radial reduction.

Spearman–Pearson gap. The pixel-space radial Pearson at `last_2` is $r = 0.337$ but Spearman is $\rho = 0.186$, a substantial gap. Roughly half the linear signal does not survive rank transformation, indicating that part of the Pearson correlation is driven by linear magnitude scaling on a heavy-tailed drift distribution rather than rank-preserving alignment. In vision-feature space the same gap closes ($r = 0.197$, $\rho = 0.197$ at `last_2`), so the issue is pixel-space-specific. We disclose this explicitly; it is a genuine caveat on the strength of the linear claim, but not a contradiction of the law, and it is absent under the 2D form and in vision-feature space.

DC handling. The predictor is highly sensitive to whether the DC bin is retained. In a paired 200-sample DC ablation (the historical $B = 20$ run on 2B), zeroing the DC bin collapses the grouped first-order r from 0.471 to 0.055, because both $W_t(\omega)$ for fine-grained tasks and natural-perturbation $\Delta F(\omega)$ concentrate 80–96% of their mass on DC and DC-adjacent bins. We retain DC by default; this is eq. (2) as written, not a tuning choice.

6 Figure Discussion

7 Discussion

Why reasoning is spectrally costly. As C_{sem} rises at the second-from-last attention layer, attention mass concentrates *downward* onto the top-1/top-2 frequency bins, while the high-frequency tail thins. This is exactly the regime where natural perturbations (noise, blur, JPEG, elastic, lighting changes) also deposit the bulk of their energy: the empirical $\Delta F(\omega)$ profile of every natural-perturbation family is dominated (80–96%) by the DC and DC-adjacent bins. The first-order overlap $\langle W_t(\omega)^{(\text{last}_2)}, \Delta F(\omega) \rangle$ therefore grows with C_{sem} , and behavioural damage follows. Fine

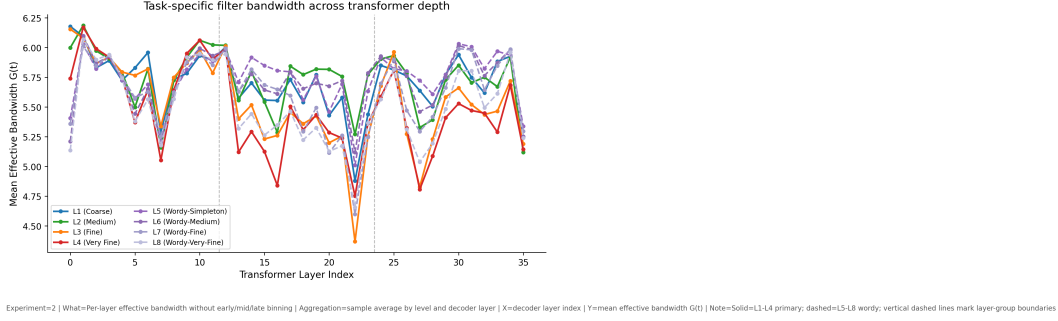


Figure 3: **Layer-resolved bandwidth and the four-stage trajectory** (Qwen3-VL-8B, 100 samples, all 36 decoder attention layers). At layer 0 the wordy mean bandwidth sits -0.74 below the primary mean (length prior). By layer 2 the gap closes; through layers 1–12 all eight levels equilibrate near bandwidth ≈ 6.0 (prior erased). Mid layers (13–23) drive C_{sem} -driven narrowing: L4 drops to $G(t) \approx 4.4$ at layer 22, while primary–wordy gap stays ≈ 0 (length-insensitive content engagement). Late layers (24–35) and `last_2` re-open the primary–wordy gap to a peak $+0.33$ at layer 31, with wordy mirrors now *broad*er than primaries (verbosity stabiliser). The bandwidth ladder by L1–L4 at `last_2` is 5.93, 5.92, 5.72, 5.68, monotone with C_{sem} . Mass concentrates downward onto DC and DC-adjacent bins, where natural perturbation $\Delta F(\omega)$ also lives—the overlap that makes fine reasoning spectrally costly.

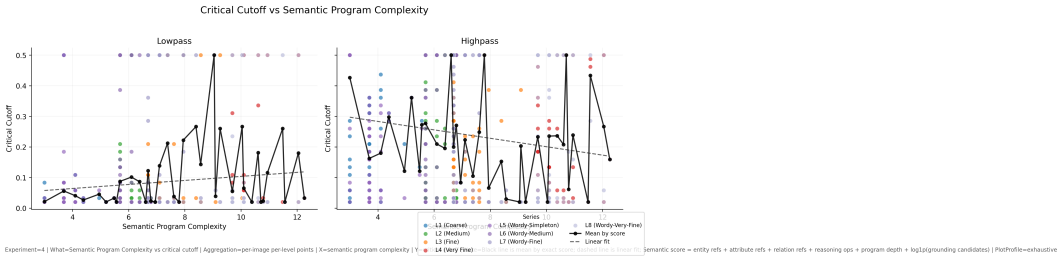


Figure 4: **Task-specific cutoff**. ω_c vs C_{sem} : fine-grained reasoning tolerates less high-frequency removal; cutoffs fall monotonically from L1 to L4, and wordy mirrors L5–L8 inherit the cutoff of their primary counterpart.

Societal impact: turning silent failure into an auditable quantity. The practical value of S_{pred} is that it gives deployment teams a pre-deployment audit tool where none existed. A hospital integrating a VLM for pre-screening radiology reports can estimate $W_t(\omega)$ for its target task on a handful of representative images, intersect it with the spectral signatures of the corruptions its imaging pipeline actually produces (motion blur, JPEG compression, anonymisation artefacts), and read off a calibrated fragility score *without* running a full behavioural sweep for every model update. An accessibility tool that describes scenes for blind users can detect in advance that a request for fine attribute reasoning on low-light photographs will be more brittle than a coarse presence query on the same input, and adjust either its prompt (padding to stabilise) or its output confidence accordingly. A safety auditor for autonomous perception has, for the first time, a parameter-free number that is estimated from the model’s own attention—not from a held-out benchmark that may or may not resemble the deployment distribution. The tug-of-war result has a second, subtler consequence: because verbose prompts *stabilise* rather than destabilise, users who phrase questions in longer, less direct language—a pattern correlated with non-native speakers, users of assistive

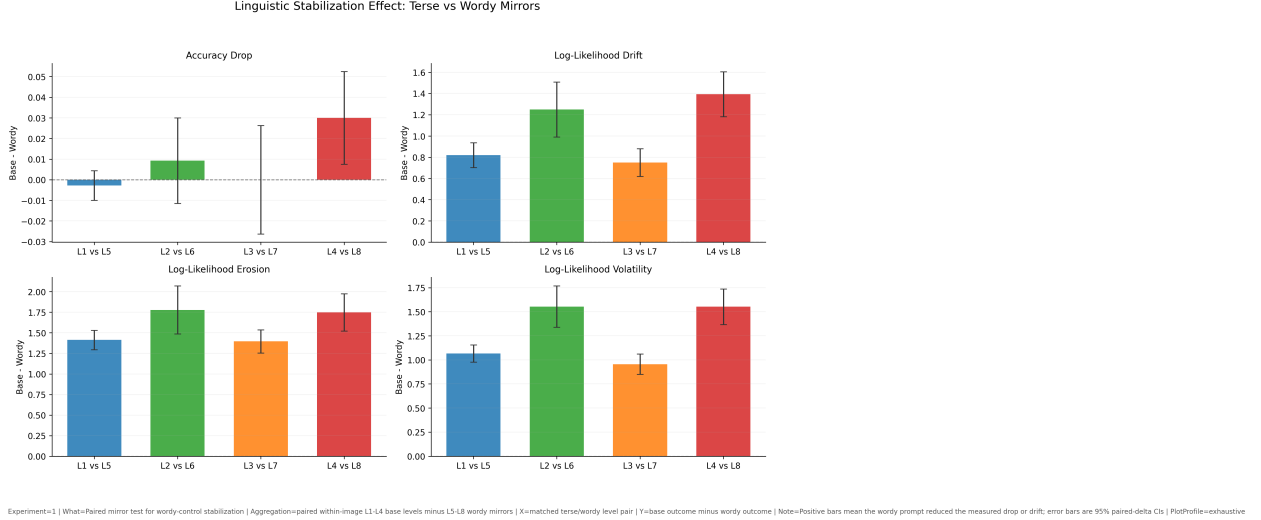


Figure 5: **Linguistic stabilisation effect across four behavioural targets.** Within-pair difference (base – wordy) for the four primary-wordy mirror pairs, on accuracy drop, log-likelihood drift, log-likelihood erosion, and log-likelihood volatility (the primary outcome). Positive bars indicate wordy mirrors stabilise. All 16 cells are positive: every wordy mirror reduces every drift measure. The effect is largest at L2↔L6 and L4↔L8 (the costliest tasks) on volatility and erosion; even on the lossy accuracy-drop target where overall variance is tiny ($R^2 = 0.02$ in the regression), the wordy direction is preserved. 95% paired-delta CIs.

dictation, and certain clinical or legal domains—may in fact receive *more* robust answers from the model than users who ask tersely. This is an encouraging finding for equitable access, and it would have been invisible to any framework that treats “prompt complexity” as a single axis. Our decomposition makes the fairness geometry explicit, and opens the door to targeted, low-cost interventions at the prompt layer rather than the expensive retraining loop.

8 Limitations

The dual-force regression uses Qwen3-VL-2B; the overlap-law and four-stage-trajectory headline numbers come from a 100-sample replication on Qwen3-VL-8B. Cross-model replication on LLaVA-NeXT [?] and InternVL [?] is ongoing. C_{sem} is equal-weighted by design; while the sign of every effect is invariant under any strictly-positive reweighting, absolute slope magnitudes are not. $W_t(\omega)$ is estimated from attention, which is a correlational lower bound on the true causal filter; activation-patching [?] remains future work. The mirror construction is synthetic; human-judged paraphrase mirrors are a natural extension. The pixel-space first-order law shows a substantial Spearman–Pearson gap (0.186 vs 0.337 at `last_2` radial) indicating heavy-tailed drift structure that the linear correlation partly captures via magnitude scaling; the gap closes in vision-feature space and under the 2D form, so the underlying law is not in question, but a fully rank-preserving variant of the predictor is open work. Finally, the radial-vs-2D trade-off is empirically informative (radial wins between-cells, 2D wins within-cells); a unified estimator that adaptively combines both is open work.

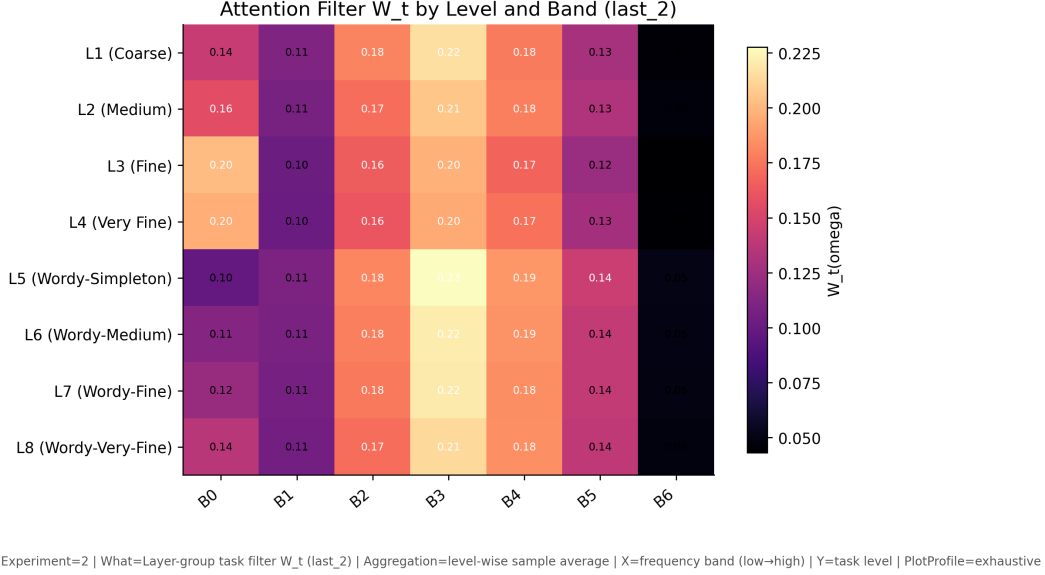


Figure 6: **Cross-modal attention spectrum $W_t(\omega)^{(\text{last}_2)}$ across the 8-level grid.** Radial-band-resolved filter mass per question level at the second-from-last attention layer ($B = 7$ auto-resolved). L1–L4 primaries (top half) concentrate mass downward onto the DC-adjacent bins as C_{sem} rises (top-2 mass grows; tail thins). L5–L8 wordy mirrors (bottom half) show a broader, more uniform mass distribution at the same depth—the verbosity-stabilising relaxation. The bandwidth ladder at `last_2` is 5.93, 5.92, 5.72, 5.68 for L1–L4 (monotone), and 5.94, 5.98, 5.99, 5.96 for L5–L8 (consistently broader).

9 Conclusion

The motivating question of this paper was practical: can we, before deploying a VLM on a given task under a given corruption profile, predict how badly it will break? Existing work can catalogue failures, show that CNNs respond to high-frequency content, or describe what cross-modal attention looks at, but none of these sources gives a practitioner a number. We give one. By recasting the conditioning text as an active spectral operator and treating reasoning as a depth-resolved trajectory of attentional mass through four distinct stages—layer-0 prompt-length prior, layer-1–12 erasure, mid-layer C_{sem} -driven narrowing (length-insensitive), and late + `last_2` content-plus-verbosity divergence—we derive a parameter-free first-order overlap predictor $\langle W_t(\omega)^{(\text{last}_2)}, \Delta F(\omega) \rangle$ that tracks behavioural damage at $r = 0.337$ pooled grouped on the 100-sample Qwen3-VL-8B replication ($r = 0.315$ per-sample, $p \approx 0$, $n = 16,800$ grouped / 62,400 sample) without any target-side fitting; the law works *within-cell* as well, with matched fixed-effects within-cell median Pearson $r = 0.543$ when the filter is restricted to question tokens. We expose a tug-of-war in which semantic logic destabilises log-likelihood volatility ($R^2 = 0.22$, $\tilde{\beta}(C_{\text{res}}) = +0.29$, $p < 10^{-130}$) while prompt length anchors it ($\tilde{\beta}(C_{\text{prompt}}) = -0.32$, $p < 10^{-209}$); every wordy mirror reduces drift variance by 65–86% relative to its primary counterpart (mean variance ratio 0.247 on 8B, reproducing the 0.246 value on 2B almost exactly), with mean log-likelihood volatility falling by a larger 68–90% on the 8B replication. We verify the tug-of-war survives even the lossy thresholding to discrete accuracy; we identify the physical mechanism as a four-stage layer trajectory in which the second-from-last attention layer’s filter concentrates downward onto the very bins where natural perturbations also concentrate; and we show that DC handling is load-bearing rather than cosmetic (paired

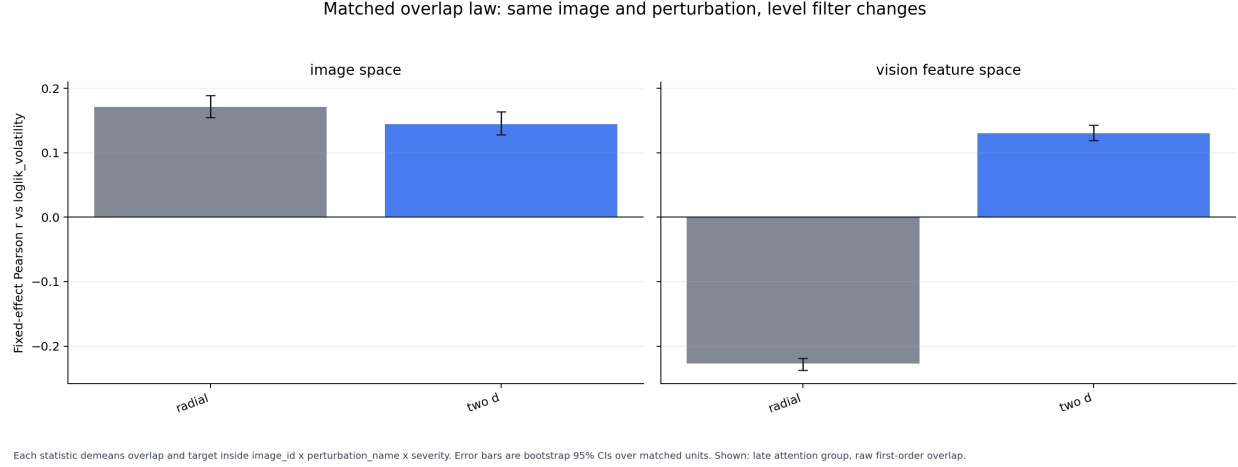


Figure 7: **Matched-FE within-cell Pearson: image vs feature space, radial vs 2D.** Within each (image $\times$ perturbation $\times$ severity) cell only the task level varies (L1–L8). Bars are the FE-Pearson correlation between within-cell demeaned overlap and within-cell demeaned log-likelihood volatility, with bootstrap 95% CIs. **Left (image-space):** both radial and 2D forms are positive and significant—the law works within-cell. **Right (vision-feature space):** the radial form is strongly negative, the 2D form is positive—the apparent “feature-space sign flip” is a radial-reduction artefact. The paper’s headline within-cell median Pearson ($r = 0.350 / 0.457 / 0.543$ in Table 5.5) uses the stronger `last_2` layer-group; this plot is the `late` third for which the matched-FE auxiliary script was run.

200-sample DC ablation collapsed $r = 0.471$ with DC on to $r = 0.055$ with DC off). We also expose a clean orientation-vs-magnitude diagnostic: the radial reduction is the cleaner aggregate predictor (between-cell radial $r = 0.337$ vs 2D $r = 0.315$) but the 2D form is the cleaner item-level predictor (within-cell median Pearson 0.457 vs 0.350), and the apparent “vision-feature sign flip” that troubled earlier drafts turns out to be a radial-reduction artefact (-0.346 radial vs $+0.193$ 2D in vision-feature space at `last_2`). The immediate payoff is a deployment audit tool—one that turns silent VLM failure into an engineerable, pre-computable quantity—and a family of cheap, training-free interventions (augmentation targeted at $W_t(\omega)$, prompt padding as a stabiliser) that follow directly from the theory. The broader payoff is that the Fourier, compositional, and mechanistic accounts of VLM fragility, which until now have talked past each other, become three facets of a single falsifiable law.

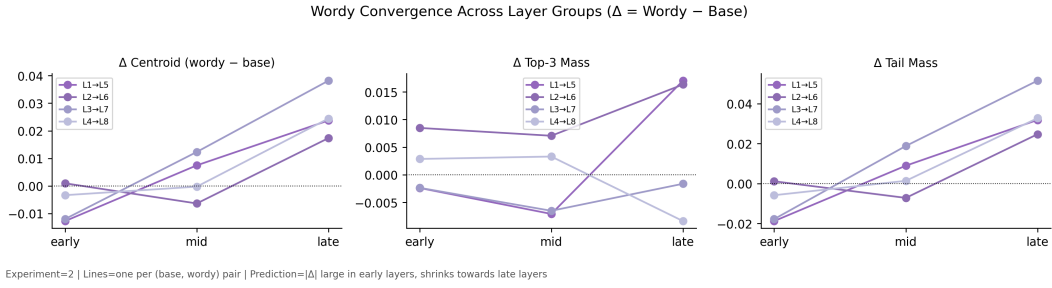


Figure 8: **Wordy–primary bandwidth gap across decoder depth** (per-layer, 100-sample 8B). The primary–wordy bandwidth difference shows the four-stage trajectory of section 5.2: a strong negative gap at layer 0 (length prior, wordy narrower); near-zero through layers 1–12 (prior erased) and the mid third 13–23 (length-insensitive content narrowing); a growing positive gap through the late third 24–35, peaking at +0.33 at layer 31 (verbosity-stabilising broadening). The sign reversal between layer 0 and layer 31 is the mechanistic signature of how prompt length is processed: applied as an immediate spectral bias, erased during semantic integration, and re-expressed in the opposite direction at deep decoder layers as a stabilising broadening.